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A DECAY ESTIMATE FOR THE FOURIER TRANSFORM OF CERTAIN SINGULAR MEASURES IN $\mathbb{R}^4$ AND APPLICATIONS

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Abstract. We consider, for a class of functions $\varphi : \mathbb{R}^2 \setminus \{\mathbf{0}\} \to \mathbb{R}^2$ satisfying a nonisotropic homogeneity condition, the Fourier transform $\widehat{\mu}$ of the Borel measure on $\mathbb{R}^4$ defined by

$$\mu(E) = \int_U \chi_E(x, \varphi(x))\, dx$$

where E is a Borel set of $\mathbb{R}^4$ and $U = \{(t^{\alpha_1}, t^{\alpha_2} s) : c < s < d,\ 0 < t < 1\}$. The aim of this article is to give a decay estimate for $\widehat{\mu}$, for the case where the set of nonelliptic points of φ is a curve in $\overline{U} \setminus \{\mathbf{0}\}$. From this estimate we obtain a restriction theorem for the usual Fourier transform to the graph of $\varphi|_U : U \to \mathbb{R}^2$. We also give L^p-improving properties for the convolution operator $T_\mu f = \mu * f$.

1. Introduction

Let U be an open bounded set in $\mathbb{R}^n$ and $\varphi : U \to \mathbb{R}^m$ be a continuous function. Let μ_U be the Borel measure on $\mathbb{R}^{n+m}$ supported on the graph of φ, defined by

$$\mu_U(E) = \int_U \chi_E(x, \varphi(x))\, dx, \tag{1}$$

where dx denotes the Lebesgue measure on $\mathbb{R}^n$. Let Σ be the graph of φ, i.e.

$$\Sigma = \{(x, \varphi(x)) : x \in U\}. \tag{2}$$

For $f \in \mathcal{S}(\mathbb{R}^{n+m})$, let $T_{\mu_U} f = \mu_U * f$ and let $\mathcal{R} f = \widehat{f}\,|_\Sigma$, where $\widehat{f}$ denotes the usual Fourier transform of f. The $L^p(\mathbb{R}^{n+m}) - L^q(\Sigma)$ boundedness of the restriction operator $\mathcal{R}$ to the submanifold $\Sigma \subset \mathbb{R}^{n+m}$ has been widely studied, in different cases, by many authors. For $S^1 \subset \mathbb{R}^2$ by C. Fefferman in [7], for $S^{n-1} \subset \mathbb{R}^n$, $n \geq 2$, by P. Tomas and E. Stein in [18] and [13] respectively, for quadratic surfaces with non vanishing gaussian curvature by R. Strichartz in [17]. The case of compact $n-$dimensional manifolds $\Sigma \subset \mathbb{R}^{2n}$ was studied by E. Prestini in [11] under the assumption that: for each $x_0 \in \Sigma$ there exists a local chart $X(x_1, ..., x_n)$ around x_0 satisfying that the vectors $\left\{\dfrac{\partial X}{\partial x_i}\right\}_{i=1}^n$ and $\left\{\dfrac{\partial^2 X}{\partial x_i \partial x_j}\right\}_{i,j=1}^n$ span $\mathbb{R}^{2n}$. The case of two dimensional manifolds $\Sigma = \big\{(x, \varphi(x)) : x \in U \subset \mathbb{R}^2\big\} \subset \mathbb{R}^4$ was studied by M. Christ in [1], where a restriction theorem is given under the assumption that: for each $x \in U$ and $\theta \in \mathbb{R}$, if $\det\left(\langle \varphi''(x), (\cos(\theta), \sin(\theta))\rangle\right) = 0$ (where $\langle\, ,\, \rangle$ denotes the inner product in $\mathbb{R}^2$) then $\frac{d}{d\theta} \det\left(\langle \varphi''(x), (\cos(\theta), \sin(\theta))\rangle\right) \neq 0$. More restriction theorems for homogeneous sub-manifolds of $\mathbb{R}^n$ can be found in [3] and [4]. A very interesting book about the Fourier restriction problem can be found in [5].

On the other hand, the $L^p(\mathbb{R}^{n+m}) - L^q(\mathbb{R}^{n+m})$ boundedness of the convolution operator T_{μ_U} has received also considerable attention in the literature. Define the type set E_{μ_U} as the set of the pairs $\left(\frac{1}{p}, \frac{1}{q}\right) \in [0,1] \times [0,1]$ such that $\|T_{\mu_U} f\|_q \leq C_{p,q} \|f\|_p$ for all $f \in \mathcal{S}(\mathbb{R}^{n+m})$, where $C_{p,q}$ is a positive constant depending only on p and q. Explicit descriptions of the set E_{μ_U} are known for many cases (see e.g. [12] and the references therein).

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In particular, S. W. Drury and K. Guo in [6] studied the case $n = m$, for $\varphi = (\varphi_1, ..., \varphi_n)$ regular enough in an open $V \subset \mathbb{R}^n$, and they proved that if each point in V is elliptic for φ and U is a bounded open set such that $\overline{U} \subset V$, then the set E_{μ_U} is the closed triangle with vertices $(0,0)$, $(1,1)$ and $\left(\frac{2}{3}, \frac{1}{3}\right)$. Elliptic points can be defined as follows: for $x \in V$, consider the quadratic form Q_x on $\mathbb{R}^n$ defined by

$$Q_x(\zeta) = \det\left(\sum_{j=1}^{n} \zeta_j \varphi_j''(x)\right), \quad \zeta = (\zeta_1, ..., \zeta_n) \in \mathbb{R}^n, \tag{3}$$

where $\varphi_j''(x) := (\mathbf{H}\varphi_j)(x)$ is the Hessian matrix of the function φ_j at x. Then, we say that $x \in V$ is elliptic if $\min_{\zeta \in S^{n-1}} |Q_x(\zeta)| > 0$, i.e.: if for all $\zeta \in S^{n-1}$ the surface $\Sigma_\zeta = \{(y, \langle \varphi(y), \zeta \rangle) : y \in V\}$ has nonzero curvature at x (although the definition of elliptic point used in [6] is different from the given above, they are equivalent, see the comments after Lemma 1 below). Similar results about the type set E_{μ_U} are given in [8] for the case when φ is a non isotropic homogeneous function such that the origin is the unique nonelliptic point.

Our aim in this paper is to study the case where $n = m = 2$ and φ is a function with non isotropic homogeneity, whose set of nonelliptic points is a curve in $\mathbb{R}^2$.

Now, we state our precise assumptions and results. Let α_1, $\alpha_2 > 0$ with $\alpha_1 \neq \alpha_2$, and for $t > 0$, $x = (x_1, x_2) \in \mathbb{R}^2$, let

$$t \bullet x = (t^{\alpha_1} x_1, t^{\alpha_2} x_2)\,. \tag{4}$$

For $a < b$, we set

$$V^{a,b} = \{t \bullet (1, s) : s \in (a, b) \text{ and } t > 0\}\,,$$

and for $a < c < d < b$, we put

$$V_1^{c,d} = \{t \bullet (1, s) : s \in (c, d) \text{ and } 0 < t < 1\}\,, \tag{5}$$

and let $\varphi = (\varphi_1, \varphi_2) : V^{a,b} \to \mathbb{R}^2$ be a function. We assume that the following hypotheses hold:

H1) φ is a real analytic function on $V^{a,b}$.

H2) For some $m \geq 3(\alpha_1 + \alpha_2)$, $\varphi(t \bullet x) = t^m \varphi(x)$ for all $x \in V^{a,b}$, and all $t > 0$.

H3) For $a < c < d < b$ and some $\sigma \in [c, d]$, the set of nonelliptic points for φ in $\overline{V^{c,d}} \setminus \{\mathbf{0}\}$ is the curve $\{t \bullet (1, \sigma) : t > 0\}$.

Under these assumptions, from Lemma 1 and Lemma 2 (see Preliminaries below), there exist two positive integers n_1 and n_2, and a positive constant D such that for δ positive and small enough

$$\min_{\zeta \in S^1} |Q_{t \bullet (1,s)}(\zeta)| \geq D\, t^d |s - \sigma|^{n_1} \text{ for all } s \in (\sigma - \delta, \sigma), \text{ and all } t > 0, \quad \text{if } \sigma \in (c, d], \tag{6}$$

and

$$\min_{\zeta \in S^1} |Q_{t \bullet (1,s)}(\zeta)| \geq D\, t^d |s - \sigma|^{n_2} \text{ for all } s \in (\sigma, \sigma + \delta), \text{ and all } t > 0, \quad \text{if } \sigma \in [c, d). \tag{7}$$

where $d = 2(m - \alpha_1 - \alpha_2)$. Taking into account the definition of elliptic points, (6) and (7) suggest to use the number $\max\{n_1, n_2\}$ as a measure of the degeneracy of ellipticity along the curve $\{t \bullet (1, \sigma) : t > 0\}$. We assume, in addition to H1 - H3, that

H4) $\max\{n_1, n_2\} < \dfrac{2m}{\alpha_1 + \alpha_2} - 3$.

Let $a < c < d < b$ such that $\sigma \in [c,d]$ (σ satisfies H3), and let μ be the measure, fixed from now on, defined by (1) taking there $U = V_1^{c,d}$, where $V_1^{c,d}$ is given by (5). Let $\widehat{\mu}$ be its Fourier transform given, for $\xi', \xi'' \in \mathbb{R}^2$, by

$$\widehat{\mu}(\xi', \xi'') = \int_{V_1^{c,d}} e^{-i\left(\langle x, \xi' \rangle + \langle \varphi(x), \xi'' \rangle\right)} \, dx.$$

In Section 3, Theorem 11, we prove that $\widehat{\mu}$ satisfies, for some constant $C > 0$, the following estimate

$$|\widehat{\mu}(\xi', \xi'')| \leq C \, |\xi''|^{-\frac{\alpha_1+\alpha_2}{m}}, \quad \text{for all } \xi' \in \mathbb{R}^2 \text{ and all } \xi'' \in \mathbb{R}^2 \setminus \{\mathbf{0}\}. \tag{8}$$

In Section 4, we consider the restriction operator $\mathcal{R}f = \widehat{f}\,|_{\Sigma}$, with Σ given by (2) (with $U = V_1^{c,d}$). Following ideas in [1], using (8) and complex interpolation for a suitable analytic family of operators, in Theorem 13, we prove that there exists a constant $C > 0$ such that $\|\mathcal{R}f\|_{L^2(\Sigma)} \leq C \, \|f\|_{L^p(\mathbb{R}^4)}$ for all $f \in \mathcal{S}(\mathbb{R}^4)$ if and only if $\dfrac{\alpha_1+\alpha_2+4m}{2(\alpha_1+\alpha_2+2m)} \leq \dfrac{1}{p} \leq 1$. Finally, using again (8), we study the type set E_μ for the convolution operator $T_\mu f = \mu * f$, for $f \in \mathcal{S}(\mathbb{R}^4)$. Theorem 14 states that, for $p = \dfrac{\alpha_1+\alpha_2+2m}{\alpha_1+\alpha_2+m}$, the closed triangle with vertices $(0,0)$, $(1,1)$ and $\left(\frac{1}{p}, \frac{1}{p'}\right)$ is contained in E_μ; moreover $\left(\frac{1}{p}, \frac{1}{p'}\right) \in \partial E_\mu$. Finally, in Section 5, we give an example of a function $\varphi = (\varphi_1, \varphi_2) : V^{a,b} \to \mathbb{R}^2$ satisfying the hypotheses H1 - H4.

2. Preliminaries

For the sequel, we consider a function $\varphi = (\varphi_1, \varphi_2) : V^{a,b} \subset \mathbb{R}^2 \setminus \{\mathbf{0}\} \to \mathbb{R}^2$ satisfying H1-H4. Given $x \in V^{a,b}$, let Q_x be the quadratic form on $\mathbb{R}^2$ defined by

$$Q_x(\zeta) = det\left(\zeta_1 \varphi_1''(x), \zeta_2 \varphi_2''(x)\right), \quad \zeta = (\zeta_1, \zeta_2) \in \mathbb{R}^2, \tag{9}$$

where, for each $j = 1, 2$, $\varphi_j''(x) := (\mathbf{H}\varphi_j)(x)$ is the Hessian matrix of the function $\varphi_j : V^{a,b} \to \mathbb{R}$ at x; and let $K(x) = (k_{ij}(x))$ be its associated symmetric matrix, i.e.: satisfying

$$Q_x(\zeta) = \langle K(x)\zeta, \zeta \rangle, \quad \zeta = (\zeta_1, \zeta_2) \in \mathbb{R}^2. \tag{10}$$

We say that a point $x \in V^{a,b}$ is elliptic for φ, if $\min\limits_{\zeta \in S^1} |Q_x(\zeta)| > 0$.

Lemma 1. *If $x \in V^{a,b}$ is a elliptic point for φ, then the eigenvalues of $K(x)$ are either both positive or both negative and*

$$\min_{\zeta \in S^1} |Q_x(\zeta)| = \min\{|\Lambda| : \Lambda \text{ is an eigenvalue of } K(x)\}.$$

Proof. Follows immediately from the definition of elliptic point and (10). □

Let us quote that a definition of elliptic point, different from the stated at the introduction, is given in [6]. Let us recall it: For $x \in V^{a,b}$, $h \in \mathbb{R}^2$, let $\varphi''(x)h$ be the 2×2 matrix whose $j-$th column is $\varphi_j''(x)h$, where $\varphi_j''(x)$ is the Hessian matrix of the function $\varphi_j : V^{a,b} \to \mathbb{R}$ at x. A point $x \in V^{a,b}$ is called elliptic in [6], if $\det(\varphi''(x)h) \neq 0$ for all $h \in \mathbb{R}^2 \setminus \{\mathbf{0}\}$. This definition es equivalent to ours. Indeed, consider the symmetric matrix $B(x)$ defined by $\langle B(x)\zeta, \zeta \rangle = \det(\varphi''(x)\zeta)$, $\zeta \in \mathbb{R}^2$. An explicit computation shows that $\det B(x) = \det K(x)$ and so the two definitions agree.

Lemma 2. *(i) There exist two homogeneous functions Λ_1 and Λ_2 on $V^{a,b}$ of degree $d = 2(m-\alpha_1-\alpha_2)$ respect to the dilations given in (4), which give the eigenvalues of $K(x)$. Moreover, Λ_1 and Λ_2 result real analytic functions on $V^{a,b}$.*
(ii) For $\delta > 0$ and small enough, it holds that:

for some $j = 1, 2$, $\min\{|\Lambda_i(x)| : i = 1, 2\} = |\Lambda_j(x)|$ *for all* $x \in V^{\sigma-\delta,\sigma}$, *if* $\sigma \in (c, d]$, *and for some* $k = 1, 2$, $\min\{|\Lambda_i(x)| : i = 1, 2\} = |\Lambda_k(x)|$ *for all* $x \in V^{\sigma,\sigma+\delta}$, *if* $\sigma \in [c, d)$.
(*iii*) *There exist two positive integers* n_1 *and* n_2, *and a positive constant* D *such that for* δ *positive and small enough*

$$\min\{|\Lambda_i(t \bullet (1, s))| : i = 1, 2\} \geq D\, t^d |s - \sigma|^{n_1} \text{ for all } s \in (\sigma - \delta, \sigma), \text{ and all } t > 0, \quad \text{if } \sigma \in (c, d]$$

and

$$\min\{|\Lambda_i(t \bullet (1, s))| : i = 1, 2\} \geq D\, t^d |s - \sigma|^{n_2} \text{ for all } s \in (\sigma, \sigma + \delta), \text{ and all } t > 0, \quad \text{if } \sigma \in [c, d),$$

with $d = 2(m - \alpha_1 - \alpha_2)$.
(*iv*) *If* $I \subset [c, d]$ *is a closed interval such that* $\sigma \notin I$, *then there exists a positive constant* D *such that* $\min\{|\Lambda_i(t \bullet (1, s))| : i = 1, 2\} \geq D\, t^d$ *for all* $s \in I$ *and all* $t > 0$.

Proof. A computation shows that the entries $k_{ij}(x)$ of $K(x)$ are given by

$$k_{11}(x) = \det \varphi_1''(x), \quad k_{22}(x) = \det \varphi_2''(x),$$

$$k_{12}(x) = k_{21}(x) = \frac{1}{2}\left(\frac{\partial^2\varphi_1}{\partial x_1^2}\frac{\partial^2\varphi_2}{\partial x_2^2} + \frac{\partial^2\varphi_1}{\partial x_2^2}\frac{\partial^2\varphi_2}{\partial x_1^2} - 2\frac{\partial^2\varphi_1}{\partial x_1 \partial x_2}\frac{\partial^2\varphi_2}{\partial x_1 \partial x_2}\right)(x).$$

From H1 it follows that the functions k_{ij}, $1 \leq i, j \leq 2$, are real analytic on $V^{a,b}$, and from H2 the k_{ij}'s result homogeneous of degree $d = 2(m-\alpha_1-\alpha_2)$, i.e.: for every $1 \leq i, j \leq 2$, $k_{ij}(t \bullet x) = t^d k_{ij}(x)$ for $x \in V^{a.b}$, and $t > 0$. It is easy to check that the eigenvalues $\Lambda_1(x)$ and $\Lambda_2(x)$ of $K(x)$ are given by

$$\Lambda_1(x) = \frac{1}{2}\left(k_{11}(x) + k_{22}(x) - \sqrt{4k_{12}^2(x) + (k_{11}(x) - k_{22}(x))^2}\right),$$

and

$$\Lambda_2(x) = \frac{1}{2}\left(k_{11}(x) + k_{22}(x) + \sqrt{4k_{12}^2(x) + (k_{11}(x) - k_{22}(x))^2}\right).$$

The homogeneity of Λ_1 and Λ_2 follow from that of the entries k_{ij}. We observe that for every $s_0 \in (a, b)$ fixed and δ positive and small enough, each $\Lambda_i(1, \cdot\,)$ is real analytic on $(s_0 - \delta, s_0 + \delta)$. Indeed, this is clear if either $k_{12}(1, s_0) \neq 0$ or $k_{11}(1, s_0) \neq k_{22}(1, s_0)$. Now, if $k_{12}(1, s_0) = 0$ and $k_{11}(1, s_0) - k_{22}(1, s_0) = 0$, we consider the function

$$f(s) = 4k_{12}^2(1, s) + (k_{11}(1, s) - k_{22}(1, s))^2. \tag{11}$$

If f is identically zero on $(s_0 - \delta, s_0 + \delta)$ for some $\delta > 0$, then $\Lambda_1(1, s) = \Lambda_2(1, s) = k_{11}(1, s)$. So that each $\Lambda_i(1, \cdot\,)$ is real analytic on $(s_0 - \delta, s_0 + \delta)$. If f is not identically zero, since $f \geq 0$ and $f(s_0) = 0$, from the analyticity of the entries k_{ij} it follows that there exist $\delta > 0$, a positive integer q, and a nonnegative real analytic function h such that $h(s_0) > 0$ and $f(s) = (s - s_0)^{2q}[h(s)]^2$ for all $s \in (s_0 - \delta, s_0 + \delta)$. Now, for $i = 1, 2$, we set

$$\widetilde{\Lambda}_i(1, s) = \frac{1}{2}\left(k_{11}(1, s) + k_{22}(1, s) + (-1)^i (s - s_0)^q h(s)\right).$$

It is clear that the functions $\widetilde{\Lambda}_i(1, \cdot\,)$ are real analytic on $(s_0 - \delta, s_0 + \delta)$. If q is even, then $\widetilde{\Lambda}_i(1, s) = \Lambda_i(1, s)$ for $s \in (s_0 - \delta, s_0 + \delta)$ and $i = 1, 2$. If q is odd, then $\widetilde{\Lambda}_1(1, s) = \Lambda_1(1, s)$ for $s \in [s_0, s_0 + \delta)$, and $\widetilde{\Lambda}_1(1, s) = \Lambda_2(1, s)$ for $s \in (s_0 - \delta, s_0)$ (similar identities are obtained for $\widetilde{\Lambda}_2(1, \cdot\,)$). Thus, the functions $x \to \widetilde{\Lambda}_i(x)$, $i = 1, 2$, give the eigenvalues of $K(x)$. From this analysis and the homogeneity of Λ_1 and Λ_2 we obtain its analyticity on $V^{a,b}$. Hence, (*i*) follows.

To see (*ii*), we observe that each point in $V^{c,\sigma}$ (resp. in $V^{\sigma,d}$) is elliptic and so, by Lemma 1, Λ_1 and Λ_2 have the same sign in $V^{c,\sigma}$ (resp. in $V^{\sigma,d}$). Then, (*ii*) follows from the fact that $\Lambda_1 \leq \Lambda_2$ on $V^{c,\sigma}$ (resp. on $V^{\sigma,d}$).

Finally, Lemma 1, (*i*) and (*ii*) allow us to get (*iii*) and (*iv*). □

Lemma 3. *Let $G : S^1 \times (a,b) \to \mathbb{R}$ be the function given by $G(\zeta, s) = \langle \varphi(1,s), \zeta \rangle$. Then,*

$$\alpha_1^2 \det\left(\zeta_1 \varphi_1''(1,s), \zeta_2 \varphi_2''(1,s)\right) = \tag{12}$$
$$m(m-\alpha_1)(GG_{ss})(\zeta,s) + \alpha_2(\alpha_1-\alpha_2)s(G_sG_{ss})(\zeta,s) - (m-\alpha_2)^2 G_s^2(\zeta,s),$$

for every $\zeta = (\zeta_1, \zeta_2) \in S^1$ and every $s \in (a,b)$.

Proof. For $\zeta \in S^1$ fixed, let $\Psi(x) = \langle \varphi(x), \zeta \rangle$, for $x \in V^{a,b}$. Clearly, the function Ψ is homogeneous of degree m respect to the dilations in (4) and so, from the Euler equations,

$$m\Psi(x) = \alpha_1 x_1 \Psi_{x_1}(x) + \alpha_2 x_2 \Psi_{x_2}(x)$$
$$(m-\alpha_1)\Psi_{x_1}(x) = \alpha_1 x_1 \Psi_{x_1x_1}(x) + \alpha_2 x_2 \Psi_{x_1x_2}(x)$$
$$(m-\alpha_2)\Psi_{x_2}(x) = \alpha_1 x_1 \Psi_{x_1x_2}(x) + \alpha_2 x_2 \Psi_{x_2x_2}(x).$$

Thus

$$mG(\zeta,s) = \alpha_1 \Psi_{x_1}(1,s) + s\alpha_2 \Psi_{x_2}(1,s) = \tag{13}$$
$$\frac{\alpha_1^2}{m-\alpha_1}\Psi_{x_1x_1}(1,s) + \frac{\alpha_2^2 s^2}{m-\alpha_2}\Psi_{x_2x_2}(1,s) + \left(\frac{\alpha_1\alpha_2}{m-\alpha_1} + \frac{\alpha_1\alpha_2}{m-\alpha_2}\right) s\, \Psi_{x_1x_2}(1,s);$$

we also have that

$$\psi_{x_2}(1,s) = G_s(\zeta,s) \quad \text{and} \quad \psi_{x_2x_2}(1,s) = G_{ss}(\zeta,s), \tag{14}$$

and so

$$(m-\alpha_2)\, G_s(\zeta,s) = \alpha_1 \Psi_{x_1x_2}(1,s) + \alpha_2\, s\, G_{ss}(\zeta,s).$$

Then

$$\Psi_{x_1x_2}(1,s) = \frac{m-\alpha_2}{\alpha_2} G_s(\zeta,s) - \frac{\alpha_2}{\alpha_1} s\, G_{ss}(\zeta,s), \tag{15}$$

from (13), using (14) and (15) we can express $\Psi_{x_1x_1}(1,s)$ in terms of $G(\zeta,s)$, $G_{ss}(\zeta,s)$ and $G_{ss}(\zeta,s)$. Taking into account this expression, and also (14) and (15), a computation of $\det(\Psi''(1,s))$ gives (12). □

Remark 4. *Let n_1, n_2, and D be as in Lemma 2. For $(\zeta,s) \in S^1 \times (a,b)$, let*

$$H(\zeta,s) = m(m-\alpha_1)(GG_{ss})(\zeta,s) + \alpha_2(\alpha_1-\alpha_2)s(G_sG_{ss})(\zeta,s) - (m-\alpha_2)^2 G_s^2(\zeta,s).$$

From (9), and Lemmas 1, 2 and 3 we have that there exists $D > 0$ such that

(i) For $\delta > 0$ and small enough, $|H(\zeta,s)| \geq D|s-\sigma|^{n_1}$ for all $s \in (\sigma-\delta, \sigma)$ and all $\zeta \in S^1$, if $\sigma \in (c,d]$.

(ii) For $\delta > 0$ and small enough, $|H(\zeta,s)| \geq D|s-\sigma|^{n_2}$ for all $s \in (\sigma, \sigma+\delta)$ and all $\zeta \in S^1$, if $\sigma \in [c,d)$.

(iii) If $I \subset [c,d] \setminus \{\sigma\}$ is a closed interval, then $|H(\zeta,s)| \geq D$ for all $s \in I$ and all $\zeta \in S^1$.

3. An estimate for $\widehat{\mu}$

For an open set $U \subset \mathbb{R}^2$, we put $\mu = \mu_U$ for the measure defined by (1). For $[c,d] \subset (a,b)$ such that $\sigma \in [c,d]$ (see hypothesis H3), we take the open set $U = V_1^{c,d}$ given by (5). Then, the change of variable $x_1 = t^{\alpha_1}$, $x_2 = st^{\alpha_2}$ gives

$$\widehat{\mu}(\xi',\xi'') = \alpha_1 \int_c^d \left(\int_0^1 e^{-i(\xi_1 t^{\alpha_1} + \xi_2 s t^{\alpha_2} + t^m \langle \varphi(1,s), \xi'' \rangle)} t^{\alpha_1+\alpha_2-1}\, dt \right) ds, \tag{16}$$

where $\xi' = (\xi_1, \xi_2)$ and $\xi'' = (\xi_3, \xi_4)$.

The following lemmas will allow us to give a decay estimate for (16).

Lemma 5. *Let $A_j \in \mathbb{R}$, for $j = 1, 2, 3$. Then there exists a positive constant C such that*

$$\left| \int_0^1 e^{-i(A_1 t^{\alpha_1} + A_2 t^{\alpha_2} + A_3 t^m)} dt \right| \leq C |A_3|^{-\frac{\alpha_1+\alpha_2}{m}}. \tag{17}$$

Proof. Without loss of generality we can assume $A_3 > 0$. The change of variable $\tau = A_3^{\frac{\alpha_1+\alpha_2}{m}} t^{\alpha_1+\alpha_2}$ allows us to express the integral in (17) as

$$\frac{1}{(\alpha_1+\alpha_2) A_3^{\frac{\alpha_1+\alpha_2}{m}}} \int_0^{A_3^{\frac{\alpha_1+\alpha_2}{m}}} e^{-i(a\tau^{\gamma_1} + b\tau^{\gamma_2} + \tau^{\gamma_3})} \, d\tau,$$

with $a := A_1 A_3^{-\frac{\alpha_1}{m}}$, $b := A_2 A_3^{-\frac{\alpha_1}{m}}$, $\gamma_j := \frac{\alpha_j}{\alpha_1+\alpha_2}$ for $j = 1, 2$ and $\gamma_3 := \frac{m}{\alpha_1+\alpha_2}$. Let $\Phi(\tau) = a\tau^{\gamma_1} + b\tau^{\gamma_2} + \tau^{\gamma_3}$. To prove the lemma it is enough to see that there exists a positive constant C independent of a, b, and A_3 such that for all $B > 1$

$$\left| \int_1^B e^{-i\Phi(\tau)} \, d\tau \right| \leq C. \tag{18}$$

For $\gamma \in \mathbb{R}$ and $j \in \mathbb{N}$, we set $\gamma(j) = \gamma(\gamma-1)\cdots(\gamma-j+1)$. We observe that $\tau^j \Phi^{(j)}(\tau) = \gamma_1(j) a \tau^{\gamma_1} + \gamma_2(j) b \tau^{\gamma_2} + \gamma_3(j)\tau^{\gamma_3}$, $j = 1, 2, 3$, and that the matrix equation

$$\begin{bmatrix} \gamma_1(1) & \gamma_1(2) & \gamma_1(3) \\ \gamma_2(1) & \gamma_2(2) & \gamma_2(3) \\ \gamma_3(1) & \gamma_3(2) & \gamma_3(3) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

has unique solution (c_1, c_2, c_3) (since $\det[\gamma_i(j)] = \gamma_1\gamma_2\gamma_3(\gamma_2-\gamma_1)(\gamma_3-\gamma_1)(\gamma_3-\gamma_2) \neq 0$). Thus $\tau^{\gamma_3} = \sum_{j=1}^3 c_j \tau^j \Phi^{(j)}(\tau)$ and so $\tau^{\gamma_3} \leq \sum_{j=1}^3 |c_j \tau^j \Phi^{(j)}(\tau)|$ for all $\tau \geq 1$. Then, $(1, B) \subset \cup_{j=1}^3 I_j$ where $I_j = \{\tau \in (1, B) : |\Phi^{(j)}(\tau)| \geq 1/3|c_j|\}$. Since $\frac{d}{d\tau}\left(\tau^{\gamma_1-\gamma_2+1}\frac{d}{d\tau}\left(\tau^{-\gamma_1+j+1}\frac{d}{d\tau}\Phi^{(j)}(\tau)\right)\right) \neq 0$ for all $\tau \geq 1$ and each $j = 1, 2, 3$, the Rolle's Theorem applied twice implies that, for each $c \in \mathbb{R}$, the equation $\Phi^{(j)}(\tau) = c$ has at most three solutions. Hence, each I_j has at most six connected components. Being $m \geq 3(\alpha_1+\alpha_2)$ we have $\gamma_3 \geq 3$, so (18) follows from the Van der Corput lemma (see e.g. [14], p. 332). □

For $\delta > 0$ and small, $\zeta \in S^1$ and $C > 0$ let

$$\begin{cases} I_1^{\delta,\zeta,C} = \left\{ s \in (\sigma-\delta, \sigma) : m(m-\alpha_1)\, G(\zeta, s)\, G_{ss}(\zeta, s) > \frac{C}{4}|s-\sigma|^{n_1} \right\}, \\ I_2^{\delta,\zeta,C} = \left\{ s \in (\sigma-\delta, \sigma) : m(m-\alpha_1)\, G(\zeta, s)\, G_{ss}(\zeta, s) < -\frac{C}{4}|s-\sigma|^{n_1} \right\}, \\ I_3^{\delta,\zeta,C} = \{ s \in (\sigma-\delta, \sigma) : (m-\alpha_2)^2\, G_s^2(\zeta, s) > \frac{C}{4}|s-\sigma|^{n_1} \}, \\ I_4^{\delta,\zeta,C} = \{ s \in (\sigma-\delta, \sigma) : \alpha_2(\alpha_1-\alpha_1)\, s\, G_s(\zeta, s)\, G_{ss}(\zeta, s) > \frac{C}{4}|s-\sigma|^{n_1} \}, \\ I_5^{\delta,\zeta,C} = \{ s \in (\sigma-\delta, \sigma) : \alpha_2(\alpha_1-\alpha_1)\, s\, G_s(\zeta, s)\, G_{ss}(\zeta, s) < -\frac{C}{4}|s-\sigma|^{n_1} \}. \end{cases} \tag{19}$$

For $\delta > 0$ and small enough and $i = 1, 2, 3, 4, 5$, let $J_i^{\delta,\zeta,C}$ be the analogous sets defined again by (19), but replacing there $(\sigma-\delta, \sigma)$ and n_1 by $(\sigma, \sigma+\delta)$ and n_2 respectively. Finally, for $\tau \in [c, d] \setminus \{\sigma\}$ such that $\sigma \notin (\tau-\delta, \tau+\delta)$ and $i = 1, 2, 3, 4, 5$, let $K_i^{\tau,\delta,\zeta,C}$ be the sets defined by (19), but replacing there $(\sigma-\delta, \sigma)$ and $\frac{C}{4}|s-\sigma|^{n_1}$ by $(\tau-\delta, \tau+\delta) \cap [c, d]$ and $\frac{C}{4}$ respectively.

Lemma 6. *(i) For every $\eta \in S^1$ there exist two positive constants ϵ_η and C_η and a neighborhood W_η of η in S^1 such that, for all $\zeta \in W_\eta$ and all $j = 1, 2, ..., 5$, the sets $I_j^{\epsilon_\eta,\zeta,C_\eta}$ and $J_j^{\epsilon_\eta,\zeta,C_\eta}$ have at most n_1 and n_2 connected components respectively.*

(ii) For every $\tau \in [c, d] \setminus \{\sigma\}$ and $\eta \in S^1$ there exist two positive constants $\epsilon_{\eta,\tau}$ and $C_{\eta,\tau}$ and a neighborhood $W_{\eta,\tau}$ of η in S^1 such that, for all $\zeta \in W_{\eta,\tau}$ and all $j = 1, ..., 5$, either $K_j^{\tau,\epsilon_{\eta,\tau},\zeta,C_{\eta,\tau}} = \emptyset$ or $K_j^{\tau,\epsilon_{\eta,\tau},\zeta,C_{\eta,\tau}} = (\tau-\epsilon_{\eta,\tau}, \tau+\epsilon_{\eta,\tau}) \cap [c, d]$.

Proof. To see (i), we proceed as follows. Given $f \in C^\infty(a,b)$, let d_f be defined by

$$d_f = \begin{cases} \min\{l \geq 0 : f^{(l)}(\sigma) \neq 0\} & \text{if } f^{(k)}(\sigma) \neq 0 \text{ for some } k \geq 0, \\ +\infty & \text{otherwise} \end{cases}.$$

For $(\zeta, s) \in S^1 \times (a,b)$, let

$$f_1(\zeta,s) = m(m-\alpha_1)\, G(\zeta,s)\, G_{ss}(\zeta,s),$$
$$f_2(\zeta,s) = (m-\alpha_2)^2\, G_s^2(\zeta,s),$$
$$f_3(\zeta,s) = \alpha_2(\alpha_1-\alpha_1)\, s\, G_s(\zeta,s)\, G_{ss}(\zeta,s),$$

and for $i = 1,2,3$ and $\eta \in S^1$ fixed, let $k_i = d_{f_i(\eta,\,\cdot\,)}$. From Remark 4 (i), we have $d_{(f_1+f_2+f_3)(\eta,\,\cdot\,)} \leq n_1$ and so $k_i \leq n_1$ for some $i = 1,2,3$. Let $A = \{i \in \{1,2,3\} : k_i \leq n_1\}$ and for $\eta \in S^1$ let $C_\eta := \min\left\{\{D\} \cup \left\{\left|\frac{1}{k_i!}\partial_s^{k_i} f_i(\eta,\sigma)\right| : i \in A\right\}\right\}$, where D is the constant in Remark 4 and ∂_s denotes the partial derivative respect to s. For $(\zeta, s) \in S^1 \times (a,b)$ let

$$F_i(\zeta,s) = f_i(\zeta,s) - \frac{C_\eta}{4}(s-\sigma)^{n_1}, \quad i = 1,2,3.$$

If $i \in A$, we have that $\partial_s^{k_i} F_i(\eta,\sigma) \neq 0$ and so there exists $\epsilon_{\eta,i} > 0$ and a neighborhood $W_{\eta,i}$ of η in S^1 such that $\partial_s^{k_i} F_i(\zeta,s) \neq 0$ for all $(\zeta,s) \in W_{\eta,i} \times (\sigma - \epsilon_{\eta,i}, \sigma)$. Then, by the Rolle's Theorem for $\zeta \in W_{\eta,i}$ and $i \in A$, the equation $f_i(\zeta,s) = \frac{C_\eta}{4}(s-\sigma)^{n_1}$ has at most k_i roots in $(\sigma - \epsilon_{\eta,i}, \sigma)$ with $k_i \leq n_1$. Now, if $i \notin A$, then $\partial_s^{n_1} F_i(\eta,\sigma) \neq 0$ and so, proceeding as above, we find also in this case $\epsilon_{\eta,i} > 0$ and $W_{\eta,i}$ such that for $\zeta \in W_{\eta,i}$, $f_i(\zeta,s) = \frac{C_\eta}{4}(s-\sigma)^{n_1}$ has at most n_1 roots in $(\sigma - \epsilon_{\eta,i}, \sigma)$. Similarly, by considering $\widetilde{F}_i(\zeta,s) = f_i(\zeta,s) + \frac{C_\eta}{4}(s-\sigma)^{n_1}$ instead of F_i, and diminishing $W_{\eta,i}$ and $\epsilon_{\eta,i}$ if necessary, we obtain that for $\zeta \in W_{\eta,i}$, $f_i(\zeta,s) = -\frac{C_\eta}{4}(s-\sigma)^{n_1}$ has at most n_1 roots in $(\sigma - \epsilon_{\eta,i}, \sigma)$. From these facts it is clear that the assertion in (i) holds for the sets $I_j^{\epsilon_\eta,\zeta,C_\eta}$ if we take $\epsilon_\eta = \min\{\epsilon_{\eta,i} : i = 1,2,3\}$ and $W_\eta = \bigcap_{i=1}^3 W_{\eta,i}$. The proof of the statement for the sets $J_j^{\epsilon_\eta,\zeta,C_\eta}$ is similar.

To see (ii), let $\tau \in [c,d] \setminus \{\sigma\}$ and let f_i be the functions defined above. From Remark 4 (iii), there exists $D > 0$ such that

$$|f_1(\zeta,\tau) + f_2(\zeta,\tau) + f_3(\zeta,\tau)| \geq D,$$

for all $\zeta \in S^1$. For $\eta \in S^1$, we define $A = \{i \in \{1,2,3\} : f_i(\eta,\tau) \neq 0\}$ and $C_{\eta,\tau} := \min\{\{D\} \cup \{|f_i(\eta,\tau)| : i \in A\}\}$. For $(\zeta,s) \in S^1 \times (a,b)$ and $i = 1,2,3$, let $F_i(\zeta,s) = f_i(\zeta,s) - \frac{C_{\eta,\tau}}{4}$ and $\widetilde{F}_i(\zeta,s) = f_i(\zeta,s) + \frac{C_{\eta,\tau}}{4}$. Thus $F_i(\eta,\tau) \neq 0$ and $\widetilde{F}_i(\eta,\tau) \neq 0$ for $i = 1,2,3$ and so there exist $\epsilon_{\eta,\tau,i}$ and a neighborhood $W_{\eta,\tau,i}$ of η in S^1 such that $F_i(\zeta,s) \neq 0$ and $\widetilde{F}_i(\zeta,s) \neq 0$ for all $\zeta \in W_{\eta,\tau,i}$ and $s \in (\tau - \epsilon_{\eta,\tau,i}, \tau + \epsilon_{\eta,\tau,i}) \cap [c,d]$. We set $W_{\eta,\tau} = \bigcap_{i=1}^3 W_{\eta,\tau,i}$ and $\epsilon_{\eta,\tau} = \min\{\epsilon_{\eta,\tau,i} : i = 1,2,3\}$. Since for $\zeta \in W_{\eta,\tau}$ and $i = 1,2,3$, the equations $f_i(\zeta,s) = \pm\frac{C_{\eta,\tau}}{4}$ have not roots in $(\tau - \epsilon_{\eta,\tau}, \tau + \epsilon_{\eta,\tau}) \cap [c,d]$ it follows, for every j, that $K_j^{\tau,\epsilon_{\eta,\tau},\zeta,C_{\eta,\tau}} = \emptyset$ or $K_j^{\tau,\epsilon_{\eta,\tau},\zeta,C_{\eta,\tau}} = (\tau - \epsilon_{\eta,\tau}, \tau + \epsilon_{\eta,\tau}) \cap [c,d]$. This completes the proof. □

Lemma 7. *Let $f \in C^1([\theta_1,\theta_2])$, $\gamma > 1$ and let I be a closed interval such that $I \subseteq [\theta_1,\theta_2]$. If one of the following two conditions holds*

$$(f'(s))^2 \geq A(s-\theta_1)^\alpha \quad \text{for all } s \in I, \tag{20}$$

$$(f'(s))^2 \geq A(\theta_2-s)^\alpha \quad \text{for all } s \in I, \tag{21}$$

for nonnegative constants A and α with $\alpha < 2\gamma - 2$, then there exists a constant $C > 0$ depending only on A, α, θ_1, θ_2 and γ such that

$$\int_I |f(s)|^{-\frac{1}{\gamma}} ds \leq C. \tag{22}$$

Proof. Let $I=[a,b]$. Assume that (20) holds. Since f' is continuous we have that either (i) $f'(s)\geq 0$ for all $s\in I$ or (ii) $f'(s)\leq 0$ for all $s\in I$. Consider the case (i). In this case either

$a)$ f is positive everywhere in $I\setminus\{\theta_1\}$,

or

$b)$ f vanishes at exactly a point $s_0\in I$.

If $a)$ holds, then for every $s\in I$

$$|f(s)|\geq\int_a^s f'(t)\,dt\geq\frac{A}{1+\frac{\alpha}{2}}\left((s-\theta_1)^{1+\frac{\alpha}{2}}-(a-\theta_1)^{1+\frac{\alpha}{2}}\right)\geq\frac{A}{1+\frac{\alpha}{2}}(s-a)^{1+\frac{\alpha}{2}}$$

which, taking into account that $\alpha<2\gamma-2$, implies (22) with the constant $C>0$ required.
If $b)$ holds, we have for $a\leq s\leq s_0$

$$|f(s)|\geq\int_s^{s_0} f'(t)\,dt\geq\frac{A}{1+\frac{\alpha}{2}}\left((s_0-\theta_1)^{1+\frac{\alpha}{2}}-(s-\theta_1)^{1+\frac{\alpha}{2}}\right)\geq\frac{A}{1+\frac{\alpha}{2}}(s_0-s)^{1+\frac{\alpha}{2}}$$

and thus $\int_a^{s_0}|f(s)|^{-\frac{1}{\gamma}}\,ds\leq C$, where C is the constant required. Similarly, one can estimate $\int_{s_0}^b|f(s)|^{-\frac{1}{\gamma}}\,ds$. So (22) holds in the case $b)$. The case (ii) reduces to (i) by considering there $-f$ instead of f. Finally, the proof of that (21) $\implies$ (22) is similar. □

Lemma 8. *Let $f\in C^2([\theta_1,\theta_2])$, $\gamma>1$ and let I be a closed interval such that $I\subseteq[\theta_1,\theta_2]$. If one of the following two conditions holds*

$$|f'(s)f''(s)|\geq A(s-\theta_1)^\alpha\quad\text{for all }\ s\in I, \tag{23}$$

$$|f'(s)f''(s)|\geq A(\theta_2-s)^\alpha\quad\text{for all }\ s\in I, \tag{24}$$

for nonnegative constants A and α with $\alpha<2\gamma-3$, then (22) holds for some $C>0$ depending only on A, α, θ_1, θ_2 and γ.

Proof. Let $I=[a,b]$. If (23) holds, we have that either (i) $\left((f')^2\right)'(t)\geq 2A(t-\theta_1)^\alpha$ for all $t\in[a,b]$ or (ii) $\left((f')^2\right)'(t)\leq -2A(t-\theta_1)^\alpha$ for all $t\in[a,b]$. If (i) holds, by integrating on $[a,s]$, we obtain

$$(f')^2(s)\geq(f')^2(a)+c_1\left((s-\theta_1)^{1+\alpha}-(a-\theta_1)^{1+\alpha}\right)\geq c_1(s-a)^{1+\alpha}\quad\text{for all}\, s\in I,$$

where c_1 is a positive constant depending only on A and α. If (ii) holds, by integrating on $[s,b]$, we get similarly as above that $(f')^2(s)\geq c_2(b-s)^{1+\alpha}$ for all $s\in I$, with $c_2=c_2(A,\alpha)$. Since $1+\alpha<2\gamma-2$, the lemma follows, in both cases (i) and (ii), from Lemma 7. Finally, the case when (24) holds is similar. □

Lemma 9. *Let $f\in C^2([\theta_1,\theta_2])$, $\gamma>1$ and let I be a closed interval such that $I\subseteq[\theta_1,\theta_2]$. If one of the following two conditions holds*

$$f(s)f''(s)\geq A(s-\theta_1)^\alpha\quad\text{for all }\ s\in I, \tag{25}$$

$$f(s)f''(s)\geq A(\theta_2-s)^\alpha\quad\text{for all }\ s\in I, \tag{26}$$

for nonnegative constants A and α with $\alpha<2\gamma-2$, then (22) holds for some $C>0$ depending only on A, α, θ_1, θ_2 and γ.

Proof. Let $I=[a,b]$. We give the proof only for the case when (25) holds, since the proof when (26) holds is similar. We assume that (25) occurs, in this case either

(i) $f\geq 0$ and $f''\geq 0$ on I,

or

(ii) $f\leq 0$ and $f''\leq 0$ on I.

If (i) holds, let s_1 be the point where the minimum of f on I is achieved. If $s_1=a$, then $f(a)\geq 0$ and $f'(a)\geq 0$. From (25), to integrate by parts, we get

$$\begin{aligned}\frac{1}{2}(f^2)'(t)&\geq f(t)f'(t)-f(a)f'(a)-\int_a^t(f')^2(r)\,dr=\int_a^t f(r)f''(r)\,dr\\&\geq c_1\left((t-\theta_1)^{1+\alpha}-(a-\theta_1)^{1+\alpha}\right)\geq c_1(t-a)^{1+\alpha},\end{aligned}$$

for a positive constant $c_1 = c_1(A,\alpha)$ and for all $t \in I$. A new integration on $[a,s]$ gives, for some positive constant $c_2 = c_2(A,\alpha)$,

$$f^2(s) \geq c_2(s-a)^{2+\alpha}, \quad \text{for all } s \in I. \tag{27}$$

Since $\alpha < 2\gamma - 2$, (27) gives (22) with a constant C as required. Now, if $s_1 = b$ we have that $f(b) > 0$ and $f'(r) \leq f'(b) \leq 0$ for every $r \in I$. Using (25), to integrate by parts on $[t,b]$, we get $-(f^2)'(t) \geq c_3(b-t)^{1+\alpha}$ for a positive constant $c_3 = c_3(A,\alpha)$ and all $t \in I$. Then, the proof in this case follows as above, but integrating on $[s,b]$. Now, we consider the case $a < s_1 < b$. This case follows to apply the two cases above on the intervals $[s_1,b]$ and $[a,s_1]$ respectively.

Finally, the case (ii) reduces to (i) by considering there $-f$ instead of f. □

Lemma 10. *Let $f \in C^2([\theta_1,\theta_2])$, $\gamma > 1$ and let I be a closed interval such that $I \subseteq [\theta_1,\theta_2]$. If one of the following two conditions holds*

$$f(s)f''(s) \leq -A(s-\theta_1)^\alpha \quad \textit{for all } s \in I, \tag{28}$$

$$f(s)f''(s) \leq -A(\theta_2-s)^\alpha \quad \textit{for all } s \in I, \tag{29}$$

for nonnegative constants A and α with $\alpha < 2\gamma - 2$, then (22) holds for some $C > 0$ depending only on A, α, θ_1, θ_2 and γ.

Proof. The proof of the case when (29) holds is analogous to that of when (28) occurs, so we will prove only that (28) $\implies$ (22). Let $I = [a,b]$. From (28) we have either (i) $f(s) > 0$ and $f''(s) < 0$ for all $s \in I \setminus \{\theta_1\}$ or (i) $f(s) < 0$ and $f''(s) > 0$ for all $s \in I \setminus \{\theta_1\}$. Consider the case (i). Let u and Φ be the functions defined on $[a,b]$ by

$$u(s) = \sin\left(\frac{(s-a)\pi}{b-a}\right),$$

and

$$\Phi(s) = A(b-a)^\alpha \pi^{-\alpha} u^\alpha(s).$$

From (28) we have that $f(a) \geq 0$, $f(b) > 0$ and $-f''(s) \geq \Phi(s)/f(s)$ for all $s \in (a,b]$. Let $\widetilde{u}$ and $\widetilde{\Phi}$ be the functions defined on $[a,b]$ by

$$\widetilde{u}(s) = u^{1+\frac{\alpha}{2}}(s),$$

and

$$\widetilde{\Phi}(s) = \frac{\pi^2(1+\frac{\alpha}{2})}{(b-a)^2}\left[u^{2+\alpha}(s) - \frac{\alpha}{2}\cos^2\left(\frac{(s-a)\pi}{b-a}\right)u^\alpha(s)\right].$$

A computation shows that $\widetilde{u}$ satisfies $-\widetilde{u}'' = \widetilde{\Phi}/\widetilde{u}$ on (a,b) and $\widetilde{u}(a) = \widetilde{u}(b) = 0$. Also, $\widetilde{\Phi} \leq \beta\Phi$ on (a,b), with $\beta = A^{-1}(b-a)^{-(2+\alpha)}\pi^{2+\alpha}(1+\frac{\alpha}{2})^2$. We claim that

$$\widetilde{u} \leq \beta^{1/2} f \quad \text{ on } (a,b). \tag{30}$$

Indeed, we have $-\widetilde{u}'' \leq \beta\Phi/\widetilde{u}$ and $-(\beta^{1/2}f)'' \geq \beta\Phi(\beta^{1/2}f)^{-1}$ on (a,b). Then

$$-(\beta^{1/2}f - \widetilde{u})'' \geq -\beta\Phi\frac{\beta^{1/2}f - \widetilde{u}}{\beta^{1/2}f\widetilde{u}} \quad \textit{ on } (a,b). \tag{31}$$

Let $I^* = \{s \in (a,b) : \beta^{1/2}f < \widetilde{u}\}$ and suppose that $I^* \neq \emptyset$. Let J be a connected component of I^*. Since $\beta^{1/2}f - \widetilde{u}$ is continuous on I^* and $\beta^{1/2}f - \widetilde{u}$ is nonnegative at a and at b, it follows that $\beta^{1/2}f - \widetilde{u}$ vanishes at ∂J, but $\Phi > 0$ on (a,b), so (31) implies that $\beta^{1/2}f - \widetilde{u}$ is strictly concave on J. Being $\beta^{1/2}f - \widetilde{u}$ negative on J, we get a contradiction. Thus (30) holds. Then

$$\int_I |f(s)|^{-\frac{1}{\gamma}}\,ds \leq \beta^{\frac{1}{2\gamma}}\int_I [\widetilde{u}(s)]^{-\frac{1}{\gamma}}\,ds = \beta^{\frac{1}{2\gamma}}(b-a)\pi^{-1}\int_0^\pi (\sin(\theta))^{-\frac{2+\alpha}{2\gamma}}\,d\theta < \infty$$

since $\alpha < 2\gamma - 2$. Finally, the case (ii) follows from the previous one applied to $-f$. □

We recall the definition of our singular measure μ in $\mathbb{R}^4$ which is supported on the graph of a real analytic function $\varphi : V^{a,b} \to \mathbb{R}^2$ satisfying H1 - H4. We consider the set $V_1^{c,d} \subset V^{a,b}$ given by (5), where $\sigma \in [c,d]$ (see hypothesis H3). Then, for every Borel set E of $\mathbb{R}^4$, we define the singular measure μ by

$$\mu(E) = \int_{V_1^{c,d}} \chi_E(x, \varphi(x))\, dx. \tag{32}$$

In the following theorem we give a decay estimate for the Fourier transform of μ.

Theorem 11. *Let μ be the singular measure given by (32). Then there exist a positive constant C such that*

$$|\widehat{\mu}(\xi', \xi'')| \leq C\, |\xi''|^{-\frac{\alpha_1+\alpha_2}{m}}, \quad \text{for all } \xi' \in \mathbb{R}^2 \text{ and all } \xi'' \in \mathbb{R}^2 \setminus \{\mathbf{0}\}. \tag{33}$$

Proof. We consider the case $\sigma \in (c,d)$; if $c = \sigma$ we work with the second inclusion in (34) below, now if $d = \sigma$, then we work with the first. From (16) and Lemma 5, we have for universal constant $C > 0$ that

$$|\widehat{\mu}(\xi', \xi'')| \leq C\, |\xi''|^{-\frac{\alpha_1+\alpha_2}{m}} \int_c^d |G(\zeta, s)|^{-\frac{\alpha_1+\alpha_2}{m}}\, ds, \quad \text{for all } (\xi', \xi'') \in \mathbb{R}^2 \times \mathbb{R}^2 \setminus \{\mathbf{0}\},$$

where $\zeta = \frac{\xi''}{|\xi''|}$. For $\eta \in S^1$, let C_η, ϵ_η and W_η be as in Lemma 6, (i). From the compactness of S^1 there exists a finite set of points $\{\eta_k\}_{k=1}^N \subset S^1$ such that $S^1 = \bigcup_{k=1}^N W_{\eta_k}$. From Remark 4, (i) and (ii), there exists $D > 0$ such that, for $\delta > 0$ and small enough,

$$(\sigma - \delta, \sigma) \subset \bigcup_{j=1}^5 I_j^{\delta,\, \zeta,\, D} \quad \text{and} \quad (\sigma, \sigma + \delta) \subset \bigcup_{j=1}^5 J_j^{\delta,\, \zeta,\, D} \quad \text{for all} \quad \zeta \in S^1, \tag{34}$$

where the sets $I_j^{\delta,\, \zeta,\, D}$ and $J_j^{\delta,\, \zeta,\, D}$ are defined by (19). Now, we pick ϵ such that $0 < \epsilon < \delta$ and $0 < \epsilon < \min\{\epsilon_{\eta_k} : 1 \leq k \leq N\}$, so $(\sigma - \epsilon, \sigma) \cup (\sigma, \sigma + \epsilon) \subset \bigcup_{j=1}^5 \left(I_j^{\epsilon,\, \zeta,\, D} \cup J_j^{\epsilon,\, \zeta,\, D} \right)$ for all $\zeta \in S^1$.

Given $\zeta \in S^1$, we have that $\zeta \in W_{\eta_k}$ for some $k = k(\zeta)$. Since $\epsilon_{\eta_{k(\zeta)}} \geq \epsilon$ and $C_{\eta_{k(\zeta)}} \leq D$, it follows that $I_j^{\epsilon,\, \zeta,\, D} \subset I_j^{\epsilon_{\eta_{k(\zeta)}},\, \zeta,\, C_{\eta_{k(\zeta)}}}$ and $J_j^{\epsilon,\, \zeta,\, D} \subset J_j^{\epsilon_{\eta_{k(\zeta)}},\, \zeta,\, C_{\eta_{k(\zeta)}}}$ for every $j = 1, 2, ..., 5$. Then

$$(\sigma - \epsilon, \sigma) \cup (\sigma, \sigma + \epsilon) \subset \bigcup_{j=1}^5 \left(I_j^{\epsilon_{\eta_{k(\zeta)}},\, \zeta,\, C_{\eta_{k(\zeta)}}} \cup J_j^{\epsilon_{\eta_{k(\zeta)}},\, \zeta,\, C_{\eta_{k(\zeta)}}} \right).$$

Hence

$$\int_{\sigma-\epsilon}^{\sigma-\epsilon} |G(\zeta, s)|^{-\frac{\alpha_1+\alpha_2}{m}}\, ds$$

$$\leq \sum_{j=1}^5 \left[\int_{I_j^{\epsilon_{\eta_{k(\zeta)}},\, \zeta,\, C_{\eta_{k(\zeta)}}}} |G(\zeta, s)|^{-\frac{\alpha_1+\alpha_2}{m}}\, ds + \int_{J_j^{\epsilon_{\eta_{k(\zeta)}},\, \zeta,\, C_{\eta_{k(\zeta)}}}} |G(\zeta, s)|^{-\frac{\alpha_1+\alpha_2}{m}}\, ds \right].$$

By Lemma 6, (i), the sets $I_j^{\epsilon_{\eta_{k(\zeta)}},\, \zeta,\, C_{\eta_{k(\zeta)}}}$ and $J_j^{\epsilon_{\eta_{k(\zeta)}},\, \zeta,\, C_{\eta_{k(\zeta)}}}$ have at most n_1 and n_2 connected components respectively, so we can estimate, uniformly on ζ, the integrals on the right-side hand of this inequality by applying Lemmas 7, 8, 9 and 10 to the function $f(s) = G(\zeta, s)$ with $\gamma = \frac{m}{\alpha_1+\alpha_2}$ and $\alpha = n_1$ or $\alpha = n_2$ according to the case.

Let $F = [c,d] \setminus (\sigma - \epsilon, \sigma + \epsilon)$, where ϵ is as above. It is clear that F is compact. For $\tau \in [c,d] \setminus (\sigma - \frac{\epsilon}{2}, \sigma + \frac{\epsilon}{2})$ and $\eta \in S^1$, let $C_{\eta,\tau}$, $\epsilon_{\eta,\tau}$ and $W_{\eta,\tau}$ be as in Lemma 6, (ii). From the compactness of S^1 there exists a finite set of points $\{\eta_k\}_{k=1}^M \subset S^1$ such that $S^1 = \bigcup_{k=1}^M W_{\eta_k,\tau}$.

From Remark 4, (iii), there exists $D > 0$ and $\delta > 0$ and small enough such that

$$(\tau - \delta, \tau + \delta) \cap [c,d] \subset \bigcup_{j=1}^5 K_j^{\tau,\delta,\zeta,D}, \text{ for all } \zeta \in S^1.$$

Let $\epsilon_\tau = \min\{\epsilon_{\eta_k,\tau} : 1 \le k \le M\}$. We take $\widetilde{\epsilon} = \min\{\delta, \epsilon_\tau\}$. Now, given $\zeta \in S^1$ we have that $\zeta \in W_{\eta_k}$ for some $k = k(\zeta)$. Since $\epsilon_{\eta_{k(\zeta)},\tau} = \epsilon_{\eta_k,\tau} \ge \widetilde{\epsilon}$ and $C_{\eta_{k(\zeta)},\tau} = C_{\eta_k,\tau} \le D$, it follows that $I_\tau := (\tau - \widetilde{\epsilon}, \tau + \widetilde{\epsilon}) \cap [c,d] \subset \bigcup_{j=1}^5 K_j^{\tau,\widetilde{\epsilon},\zeta,D} \subset \bigcup_{j=1}^5 K_j^{\tau,\epsilon_{\eta_k},\tau,\zeta,C_{\eta_k,\tau}}$. Thus

$$\int_{I_\tau} |G(\zeta,s)|^{-\frac{\alpha_1+\alpha_2}{m}}\, ds \le \sum_{j=1}^5 \int_{K_j^{\tau,\epsilon_{\eta_k},\tau},\,\zeta,\,C_{\eta_k,\tau}} |G(\zeta,s)|^{-\frac{\alpha_1+\alpha_2}{m}}\, ds.$$

By Lemma 6, (ii), each $K_j^{\tau,\epsilon_{\eta_k},\tau},\,\zeta,\,C_{\eta_k,\tau}$ is either empty or connected and so the integrals on the right-side hand of this last inequality can be estimated, uniformly on ζ, by using Lemmas 7, 8, 9 and 10. Since F is covered by a finite number of these intervals $(\tau-\widetilde{\epsilon},\tau+\widetilde{\epsilon})$, we obtain an estimate, uniformly on ζ, for the integral $\int_F |G(\zeta,s)|^{-\frac{\alpha_1+\alpha_2}{m}}\, ds$. This concludes the proof of the theorem. □

4. A restriction theorem for the Fourier transform and L^p-improving properties of μ

Let $\Sigma = \Big\{(x,\varphi(x)) : x \in V_1^{c,d}\Big\}$, with $\sigma \in [c,d]$ (see hyp. H3). For $f \in L^1(\mathbb{R}^4)$, let $\mathcal{R}f = \widehat{f}|_\Sigma$ where $\widehat{f}$ denotes the usual Fourier transform. The following lemma gives a necessary condition for the $L^p(\mathbb{R}^4) - L^q(\Sigma)$ boundedness of the restriction operator $\mathcal{R}$ and it is an adaptation, to our setting, of the well known Knapp's homogeneity argument.

Lemma 12. *If there exists a positive constant C such that*

$$\|\mathcal{R}f\|_{L^q(\Sigma,d\mu)} \le C\|f\|_{L^p(\mathbb{R}^4)}, \quad f \in \mathcal{S}(\mathbb{R}^4). \tag{35}$$

then $\frac{1}{q} \ge \frac{(\alpha_1+\alpha_2+2m)}{(\alpha_1+\alpha_2)}\frac{1}{p'}$.

Proof. We observe that if (35) holds for $f \in \mathcal{S}(\mathbb{R}^4)$, then it holds also for $f \in L^1(\mathbb{R}^4) \cap L^p(\mathbb{R}^4)$. Let $Q = \Big\{t \bullet (1,s) : s \in [s_1,s_2] \text{ and } \frac{1}{2} \le t \le 1\Big\}$, where $c < s_1 < s_2 < d$. We fix $0 < \eta < 1$ such that $|\varphi(x)| < 1$ for all $x \in \eta \bullet Q$. For $\xi', \xi'' \in \mathbb{R}^2$ and $t > 0$ let $t \odot (\xi', \xi'') = (t \bullet \xi', t^m \xi'')$, and for $0 < \epsilon < 1$ we set $E_\epsilon = \epsilon^{-1} \odot [0,1]^4$ and $f_\epsilon = \chi_{E_\epsilon}$.

For $x = (x_1, x_2) \in Q$, a change of variable gives

$$\widehat{f_\epsilon}(x,\varphi(x)) = \int_{E_\epsilon} e^{-i(x_1\xi_1 + x_2\xi_2 + \xi_3\varphi_1(x) + \xi_4\varphi_2(x))}\, d\xi$$

$$= \epsilon^{-(\alpha_1+\alpha_2+2m)} \int_{[0,1]^4} e^{-i(\epsilon^{-\alpha_1}x_1\xi_1 + \epsilon^{-\alpha_2}x_2\xi_2 + \epsilon^{-m}\xi_3\varphi_1(x) + \epsilon^{-m}\xi_4\varphi_2(x))}\, d\xi,$$

where $d\xi = d\xi_1 \cdots d\xi_4$. We define $F_\epsilon = (\epsilon\eta) \bullet Q$ and $\Sigma_\epsilon = \{(x,\varphi(x)) : x \in F_\epsilon\}$. Then,

$$\|\mathcal{R}f_\epsilon\|^q_{L^q(\Sigma,d\mu)} \ge \|\mathcal{R}f_\epsilon\|^q_{L^q(\Sigma_\epsilon,d\mu)}$$

$$\ge \epsilon^{-q(\alpha_1+\alpha_2+2m)} \int_{F_\epsilon} \left| \int_{[0,1]^4} e^{-i(\epsilon^{-\alpha_1}x_1\xi_1 + \epsilon^{-\alpha_2}x_2\xi_2 + \epsilon^{-m}\xi_3\varphi_1(x) + \epsilon^{-m}\xi_4\varphi_2(x))}\, d\xi \right|^q dx.$$

Also

$$\int_{[0,1]^4} e^{-i(\epsilon^{-\alpha_1}x_1\xi_1 + \epsilon^{-\alpha_2}x_2\xi_2 + \epsilon^{-m}\xi_3\varphi_1(x) + \epsilon^{-m}\xi_4\varphi_2(x))}\, d\xi$$

$$= \int_0^1 e^{-i\epsilon^{-\alpha_1}x_1\xi_1} d\xi_1 \int_0^1 e^{-i\epsilon^{-\alpha_2}x_2\xi_2} d\xi_2 \int_0^1 e^{-i\epsilon^{-m}\xi_3\varphi_1(x)} d\xi_3 \int_0^1 e^{-i\epsilon^{-m}\xi_4\varphi_2(x)} d\xi_4.$$

For $x \in F_\epsilon$ we have that $|\varphi_2(\epsilon^{-1} \bullet x)| < 1$, so the last integral in the product above can be lower bounded as

$$\left| \int_0^1 e^{-i\epsilon^{-m}\xi_4\varphi_2(x)} d\xi_4 \right| = \left| \frac{e^{-i\varphi_2(\epsilon^{-1}\bullet x)} - 1}{\varphi_2(\epsilon^{-1} \bullet x)} \right| \ge \frac{\sin(\varphi_2(\epsilon^{-1} \bullet x))}{\varphi_2(\epsilon^{-1} \bullet x)} \ge \sin(1).$$

The others integrals can be estimated similarly. Thus

$$\|\mathcal{R}f_\epsilon\|_{L^q(\Sigma,d\mu)} \ge C\epsilon^{-(\alpha_1+\alpha_2+2m)+\frac{\alpha_1+\alpha_2}{q}}, \tag{36}$$

with $C>0$ independent of ϵ. On the other hand $\|f_\epsilon\|_{L^p(\mathbb{R}^4)}=\epsilon^{-\frac{\alpha_1+\alpha_2+2m}{p}}$, so (35) and (36) lead to

$$C\epsilon^{-(\alpha_1+\alpha_2+2m)+\frac{\alpha_1+\alpha_2}{q}}\leq \epsilon^{-\frac{\alpha_1+\alpha_2+2m}{p}},$$

this implies, taking ϵ small enough, that $\frac{1}{q}\geq \frac{(\alpha_1+\alpha_2+2m)}{(\alpha_1+\alpha_2)}\frac{1}{p'}$. □

Theorem 13. *There exists a positive constant C such that*

$$\|\mathcal{R}f\|_{L^2(\Sigma,d\mu)}\leq C\|f\|_{L^p(\mathbb{R}^4)},\quad f\in\mathcal{S}(\mathbb{R}^4) \tag{37}$$

if and only if $\frac{\alpha_1+\alpha_2+4m}{2(\alpha_1+\alpha_2+2m)}\leq\frac{1}{p}\leq 1$.

Proof. We consider, for z in the strip $-\frac{\alpha_1+\alpha_2}{m}\leq\Re(z)\leq 2$, the analytic family of distributions I_z on $\mathbb{R}^2$ by

$$I_z(t)=\frac{2^{z-1}}{\Gamma\left(\frac{z}{2}\right)}|t|^{z-2}\quad\text{for }\Re(z)>0,$$

and by its analytic continuation for $\Re(z)\leq 0$ (see [9]). Let J_z be the distribution on $\mathbb{R}^4$ defined by

$$J_z=\delta_{\mathbb{R}^2}\otimes I_z\quad\text{for }-\frac{\alpha_1+\alpha_2}{m}\leq\Re(z)\leq 2, \tag{38}$$

where $\delta_{\mathbb{R}^2}$ is the Dirac distribution on $\mathbb{R}^2$ supported at the origin. Let $T_zf=\widehat{(J_z*\mu)}*f$, for $f\in\mathcal{S}(\mathbb{R}^4)$. A computation gives that

$$\|J_z*\mu\|_{L^\infty(\mathbb{R}^4)}\leq\left|\frac{2^{z-1}}{\Gamma\left(\frac{z}{2}\right)}\right|\quad\text{for }\Re(z)=2,$$

so $\|T_zf\|_{L^2(\mathbb{R}^4)}=\left\|\widehat{T_zf}\right\|_{L^2(\mathbb{R}^4)}\leq C_z\left\|\widehat{f}\right\|_{L^2(\mathbb{R}^4)}=C_z\|f\|_{L^2(\mathbb{R}^4)}$ for $\Re(z)=2$. On the other hand, one also can see that

$$\left|\widehat{J_z}(\xi)\right|\leq C\left|\xi''\right|^{\frac{\alpha_1+\alpha_2}{m}}\quad\text{for }\Re(z)=-\frac{\alpha_1+\alpha_2}{m}.$$

So, Young's inequality and Theorem 11 give $\|T_zf\|_{L^\infty(\mathbb{R}^4)}=\|(\widehat{J_z}\widehat{\mu})*f\|_{L^\infty(\mathbb{R}^4)}\leq C\left|\frac{2^{1-z}}{\Gamma\left(1-\frac{z}{2}\right)}\right|\|f\|_{L^1(\mathbb{R}^4)}$, where C is independent of such z. Moreover, it is easy to check, by means of Stirling formula (see e.g. [16]), that the family $\{T_z\}$ satisfies the hypotheses of the Stein's complex interpolation theorem (see [15], pp. 205), thus $\|T_0\|_{L^p(\mathbb{R}^4)\to L^{p'}(\mathbb{R}^4)}<+\infty$ for $p=\frac{2(\alpha_1+\alpha_2+2m)}{\alpha_1+\alpha_2+4m}$. This implies (37) for such p (see [17], Lemma 1). Finally, it is immediate to see that $\mathcal{R}:L^1(\mathbb{R}^4)\to L^2(\Sigma,\mu)$ is bounded, so the theorem follows from the Riesz-Thorin Convexity Theorem and Lemma 12. □

Let μ be a Borel measure and let T_μ be the convolution operator by μ, defined by $T_\mu f=\mu*f$, $f\in\mathcal{S}(\mathbb{R}^4)$, and let E_μ be the set of all pairs $\left(\frac{1}{p},\frac{1}{q}\right)\in[0,1]\times[0,1]$ such that $\|T_\mu f\|_{L^q(\mathbb{R}^4)}\leq C\|f\|_{L^p(\mathbb{R}^4)}$ for all $f\in\mathcal{S}(\mathbb{R}^4)$ and for some positive constant C depending only on p and q. We say that the measure μ is L^p-improving if E_μ does not reduce to the diagonal $\frac{1}{p}=\frac{1}{q}$. The following theorem states that the measure μ given by (32) is L^p-improving.

Theorem 14. *For $p=\frac{\alpha_1+\alpha_2+2m}{\alpha_1+\alpha_2+m}$, the closed triangle with vertices $(0,0)$, $(1,1)$ and $\left(\frac{1}{p},\frac{1}{p'}\right)$ is contained in E_μ and $\left(\frac{1}{p},\frac{1}{p'}\right)\in\partial E_\mu$.*

Proof. It is known that if $\left(\frac{1}{p},\frac{1}{q}\right)\in E_\mu$, then $\frac{1}{q}\leq\frac{1}{p}$ and $\frac{1}{q}\geq\frac{2}{p}-1$ (cf. [15], pp. 33; and [10], Theorem 1). Moreover, H1 implies that $\frac{1}{q}\geq\frac{1}{p}-\frac{\alpha_1+\alpha_2}{\alpha_1+\alpha_2+2m}$ (cf. [12], Proposizione 2.2, pp. 243). In particular, these facts imply that if $\left(\frac{1}{p},\frac{1}{p'}\right)\in E_\mu$, then $\frac{1}{2}\leq\frac{1}{p}\leq\frac{\alpha_1+\alpha_2+m}{\alpha_1+\alpha_2+2m}$. On the other hand, by Riesz-Thorin Theorem, E_μ is a convex set and, since μ is finite, the diagonal $\frac{1}{q}=\frac{1}{p}$ is contained in E_μ. Thus, to prove the theorem it is enough to see that $\left(\frac{1}{p},\frac{1}{p'}\right)\in E_\mu$ for $p=\frac{\alpha_1+\alpha_2+2m}{\alpha_1+\alpha_2+m}$.

Consider now, for $z \in \mathbb{C}$, the analytic family of distributions J_z defined by (38). For z in the strip $-\frac{\alpha_1+\alpha_2}{m} \leq \Re(z) \leq 2$, let S_z be the analytic family of operators defined by

$$S_z f = \mu * J_z * f, \quad f \in \mathcal{S}(\mathbb{R}^4).$$

From Theorem 11, we have $|\widehat{\mu}(\xi', \xi'')| \leq C|\xi''|^{-\frac{\alpha_1+\alpha_2}{m}}$, and since $\widehat{J_z} = 1 \otimes I_{2-z}$ we obtain that

$$\|S_z\|_{L^2(\mathbb{R}^4)\to L^2(\mathbb{R}^4)} \leq \left\|\widehat{\mu}\widehat{J_z}\right\|_\infty \leq C \left|\frac{2^{1-z}}{\Gamma\left(1-\frac{z}{2}\right)}\right| \quad \text{for } \Re(z) = -\frac{\alpha_1+\alpha_2}{m},$$

where C is independent of such z. On the other hand, it is easy to check that $(\mu * J_z)(x', x'') = \frac{2^{z-1}}{\Gamma\left(\frac{z}{2}\right)}|x'' - \varphi(x')|^{z-2}$ for $x', x'' \in \mathbb{R}^2$ and so

$$\|S_z\|_{L^1(\mathbb{R}^4)\to L^\infty(\mathbb{R}^4)} = \|\mu * J_z\|_\infty \leq \left|\frac{2^{z-1}}{\Gamma\left(\frac{z}{2}\right)}\right| \quad \text{for } \Re(z) = 2.$$

As in Theorem 13, the family $\{S_z : -\frac{\alpha_1+\alpha_2}{m} \leq \Re(z) \leq 2\}$ satisfies the hypotheses of the complex interpolation theorem and then, since $-\frac{\alpha_1+\alpha_2}{m}t + 2(1-t) = 0$ for $t = \frac{2m}{\alpha_1+\alpha_2+2m}$, S_0 is a bounded operator from $L^p(\mathbb{R}^4)$ into $L^{p'}(\mathbb{R}^4)$ for $\frac{1}{p} = \frac{t}{2} + (1-t) = \frac{\alpha_1+\alpha_2+m}{\alpha_1+\alpha_2+2m}$. Finally, being T_μ a non-zero multiple of S_0, the theorem follows. □

Remark 15. *If one consider $\varphi \in C^\infty(V^{a,b})$ with additional hypotheses in the lemma 2, then the theorems 13 and 14 are still true. More precisely, let $K(x) = [k_{ij}(x)]$ be the symmetric matrix associated to the quadratic form in (9). If we assume that for every $s_0 \in (a,b)$ there exists a natural number $r = r(s_0)$ such that $\left[\frac{d^r}{ds^r}k_{ij}(1,s)\right]_{s=s_0} \neq 0$, then the function f appearing in (11) is such that $\left[\frac{d^m}{ds^m}f(s)\right]_{s=s_0} \neq 0$ for some $m \geq 1$. Indeed, if we assume the hypothesis above, then $f = g^2 + h^2$ with $\left[\frac{d^n}{ds^n}g(s)\right]_{s=s_0} \neq 0$ and $\left[\frac{d^m}{ds^m}h(s)\right]_{s=s_0} \neq 0$, where n and m are the smallest natural with this property, without loss of generality we can assume $f = g^2$, applying the Leibniz formula we get $\frac{d^{2n}}{ds^{2n}}f(s) = \frac{d^{2n}}{ds^{2n}}g^2(s) = C_n\left(\frac{d^n}{ds^n}g(s)\right)^2$, where C_n depends only on n; so f has the property required. Hence, the lemma 2 still holds if we replace the word analytic by smooth and we consider an homogeneous function $\varphi = (\varphi_1, \varphi_2) \in C^\infty(V^{a,b})$ such that for each $s_0 \in (a,b)$ there exist two natural numbers $n = n(s_0)$ and $m = m(s_0)$ for which $\left[\frac{\partial^n}{\partial s^n}\varphi_1(1,s)\right]_{s=s_0} \neq 0$ and $\left[\frac{\partial^m}{\partial s^m}\varphi_2(1,s)\right]_{s=s_0} \neq 0$. Finally, we observe that the remain of the lemmas hold considering only $\varphi \in C^\infty(V^{a,b})$.*

5. An example

Let $\varphi : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

$$\varphi(x,y) = (\varphi_1(x,y), \varphi_2(x,y)) = \left(\frac{-4}{132}x^{12} + \frac{1}{30}y^6,\ \frac{-3}{132}x^{12} + \frac{1}{30}y^6 + \frac{5}{90}x^6y^3\right), \quad (x,y) \in \mathbb{R}^2.$$

In this case we have that $\varphi(t \bullet (x,y)) = t^m\varphi(x,y)$ with $\alpha_1 = \frac{1}{2}$, $\alpha_2 = 1$ and $m = 6$, so H1 and H2 hold. A computation shows that the discriminant of the quadratic form

$$(\zeta_1, \zeta_2) \to \det\left(\zeta_1\varphi_1''(1,y) + \zeta_1\varphi_2''(1,y)\right) = \langle K(1,y)\,\zeta,\, \zeta\rangle$$

is

$$\det K(1,y) = -4y^4\left(-3 + \frac{5}{3}y^3\right)\left(y^4 + \frac{y}{3}\right) + 4y^8 - \frac{1}{4}\left[4\left(y^4 + \frac{y}{3}\right) + y^4\left(3 - \frac{5}{3}y^3\right)\right]^2.$$

We observe that $k_{11}(1,y) = -4y^4$, and the discriminant is positive for every $y \in \left[\frac{97}{100}, 1\right)$ and it has a simple zero for $y = 1$. Since $\Lambda_1(1,1)\Lambda_2(1,1) = \det K(1,1) = 0$ and $\left[\frac{d}{dy}\det K(1,y)\right]_{y=1} \neq 0$, it follows that $\Lambda_1(1,1) \neq 0$ and $\left[\frac{d}{dy}\Lambda_2(1,y)\right]_{y=1} \neq 0$ or $\Lambda_2(1,1) \neq 0$ and $\left[\frac{d}{dy}\Lambda_1(1,y)\right]_{y=1} \neq 0$. Thus, H3 holds with $\sigma = 1$. It is clear that H4 also holds. Finally H1 - H4 hold for φ and $V = \left\{(t^{1/2}, ty) : \frac{97}{99} < y < 1 \,\text{and}\, 0 < t < 1\right\}$.

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